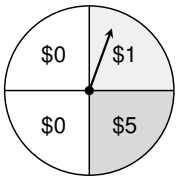
Unit 7 Review KEY

Name _____

One game, every tool. A game booth at the school fair raises money. You pay $2 to spin this spinner one time. It has 4 equal parts. You win the prize on your part.

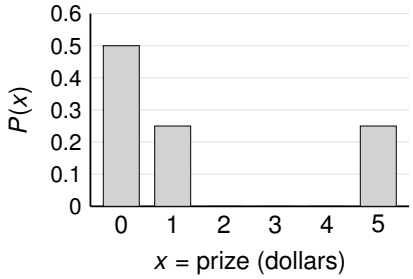


7.1 Make the distribution.
List every value of X with its probability.
Each $P(x)$ is from 0 to 1.
$\Sigma P(x) = 1$
(Σ means add up.)

X = the prize from one spin, in dollars.

x	0	1	5
$P(x)$	$\frac{2}{4} = \frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

Check: $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$. Valid.



7.1 Theoretical or empirical?
Theoretical: from equally likely outcomes.
Empirical: from data.
Divide each count by the total.

The 4 parts are equally likely, so the table above is **theoretical**.
The booth counted 100 spins: $0 52 times, $1 23 times, $5 25 times.
This is the **empirical** distribution:
$P(0) = \frac{52}{100} = 0.52$, $P(1) = 0.23$, $P(5) = 0.25$
It is close to the theoretical one, but not the same.

7.2 Find the expected value.
$E(X) = \Sigma\, x \cdot P(x)$
Multiply each value by its probability. Then add.
$E(X)$ is also the mean of X, written μ_X.

x	0	1	5
$P(x)$	0.5	0.25	0.25
$x \cdot P(x)$	0	0.25	1.25

$E(X) = 0 + 0.25 + 1.25 = \$1.50$

Over many spins, a spin pays about $1.50 on average. No single spin pays $1.50.

7.3 Find the expected payoff.
Net gain = what you win − what you paid.
A game is **fair** if the expected payoff is 0.

A spin costs $2.

Outcome	Net gain x	$P(x)$	$x \cdot P(x)$
Win $0	$0 - 2 = -2$	0.5	-1
Win $1	$1 - 2 = -1$	0.25	-0.25
Win $5	$5 - 2 = 3$	0.25	0.75

Expected payoff = $-1 - 0.25 + 0.75 = -\$0.50$
On average, a player loses 50 cents a spin. The game is **not fair**. The booth keeps that money for the school. A spin that cost $1.50 would be fair.

Expected value is a long-run average. It tells you what happens over many tries. It does not tell you what happens on one try. For a decision, also think about risk: can you pay for the worst outcome?

Work with a partner. Compare answers when you finish each problem.

7.1 - Probability Distributions

1. **I check if a probability distribution is valid, and I draw it as a graph.**

   Is each table a valid probability distribution? Explain.

(a)

x	0	1	2
$P(x)$	0.25	0.45	0.30

Valid. Each one is from 0 to 1, and $0.25 + 0.45 + 0.30 = 1$.

(b)

x	1	2	3	4
$P(x)$	0.1	0.3	0.3	0.2

Not valid. $0.1 + 0.3 + 0.3 + 0.2 = 0.9$, not 1.

(c)

x	0	1	2
$P(x)$	0.6	0.5	-0.1

Not valid. They add to 1, but -0.1 is less than 0.

(d)

x	1	2	3
$P(x)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

Valid. $\frac{3}{6} + \frac{2}{6} + \frac{1}{6} = \frac{6}{6} = 1$.

Let X = the number of library books a student checks out in one visit.

x	0	1	2	3	4
$P(x)$	0.10	0.40	?	0.15	0.05

(e) Find the missing probability, $P(2)$.

$1 - (0.10 + 0.40 + 0.15 + 0.05)$
$= 1 - 0.70 = 0.30$

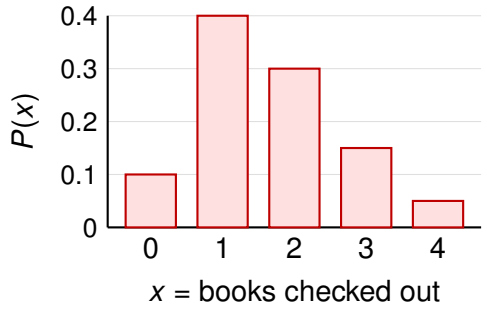
(f) Find $P(X \geq 2)$.

$0.30 + 0.15 + 0.05 = 0.50$

(g) Find $P(1 \leq X \leq 3)$.
($\leq$ means "less than or equal to.")

$0.40 + 0.30 + 0.15 = 0.85$

(h) Draw a bar graph of the distribution.



(i) Find the missing probability.

x	1	2	3	4
$P(x)$	$\frac{1}{8}$	$\frac{3}{8}$	?	$\frac{1}{4}$

$1 - \left(\frac{1}{8} + \frac{3}{8} + \frac{2}{8}\right) = \frac{2}{8} = \frac{1}{4}$

2. **I explain what a random variable is, and I make theoretical and empirical distributions.**

Is each one a random variable? Explain.

(a) The number of heads when you flip a coin 4 times.
Yes. It is a number, and it comes from a chance process.

(b) The number of days in the month of May.
No. It is always 31. It does not come from chance.

(c) The color of the next car that drives past the school.
No. It comes from chance, but it is not a number.

A spinner has 3 equal parts, numbered 1, 2, and 3. Spin it 2 times. Let X = the sum of the two numbers.

(d) List all 9 outcomes and the sum for each.
1+1 = 2, 1+2 = 3, 1+3 = 4, 2+1 = 3, 2+2 = 4, 2+3 = 5, 3+1 = 4, 3+2 = 5, 3+3 = 6

(e) Make the theoretical probability distribution of X.

x	2	3	4	5	6
$P(x)$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{3}{9}$	$\frac{2}{9}$	$\frac{1}{9}$

(f) Find $P(X \geq 5)$. $\frac{2}{9} + \frac{1}{9} = \frac{3}{9} = \frac{1}{3}$

A class did this 100 times. Each time, they spun 2 times and wrote the sum.

Sum	2	3	4	5	6
Count	9	24	30	25	12

(g) Make the empirical probability distribution of X.

x	2	3	4	5	6
$P(x)$	0.09	0.24	0.30	0.25	0.12

(h) Compare $P(4)$ in the two distributions. Why are they different?
Theoretical: $\frac{3}{9} \approx 0.33$. Empirical: 0.30. The empirical one comes from only 100 tries. Chance makes the counts vary.

(i) The class does this 1,000 more times. What do you expect to happen to the empirical distribution?
It should get closer to the theoretical distribution. More trials give results closer to the theoretical probabilities.

7.2 - Expected Value

3. **I find the expected value of a random variable, and I explain what it means.**

Let X = the number of hits a baseball player gets in one game.

x (hits)	0	1	2	3
$P(x)$	0.25	0.45	0.20	0.10
$x \cdot P(x)$	0	0.45	0.40	0.30

(a) Fill in the last row of the table.

(b) Find $E(X)$.
$E(X) = \Sigma\, x \cdot P(x) = 0 + 0.45 + 0.40 + 0.30 = 1.15$ hits

(c) What does $E(X)$ mean?
Over many games, the player gets about 1.15 hits per game, on average.

(d) A student says, "The expected value must be wrong. A player cannot get part of a hit." Explain the mistake.
The expected value is a long-run average. It does not have to be a value that can happen in one game.

(e) The player plays 40 games. About how many hits do you expect? $40 \times 1.15 = 46$ hits

Flip a coin 4 times. Let X = the number of heads.

x	0	1	2	3	4
$P(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

(f) Find $E(X)$.
$0 \cdot \frac{1}{16} + 1 \cdot \frac{4}{16} + 2 \cdot \frac{6}{16} + 3 \cdot \frac{4}{16} + 4 \cdot \frac{1}{16} = \frac{0+4+12+12+4}{16} = \frac{32}{16} = 2$ heads

(g) This distribution is symmetric. How does that help you check your answer to (f)?
The probabilities are the same on both sides of 2. So the mean is in the middle, at 2.

(h) Another name for $E(X)$ is μ_X, the mean of X. (μ is "mu.") Find μ_X for this table.

x	10	20	50
$P(x)$	0.5	0.3	0.2

$\mu_X = 10(0.5) + 20(0.3) + 50(0.2) = 5 + 6 + 10 = 21$

4. **I decide if a method of choosing is fair, and I decide if a game is fair.**

Rosa, Tam, and Yusuf need to choose who goes first in a game. Is each method fair? Explain.

(a) Roll a number cube. 1 or 4: Rosa. 2 or 5: Tam. 3 or 6: Yusuf.
Fair. Each person has $\frac{2}{6} = \frac{1}{3}$.

(b) Draw one card from a standard deck. Hearts: Rosa. Diamonds: Tam. Clubs or spades: Yusuf.
Not fair. Rosa $\frac{13}{52} = \frac{1}{4}$, Tam $\frac{13}{52} = \frac{1}{4}$, Yusuf $\frac{26}{52} = \frac{1}{2}$.

(c) Draw one card from a standard deck. A face card (J, Q, K): Rosa. An ace: Tam. Any other card: Yusuf.
Not fair. Rosa $\frac{12}{52}$, Tam $\frac{4}{52}$, Yusuf $\frac{36}{52}$.

(d) Flip a coin. Heads: Rosa. Tails: roll a number cube. 1, 2, or 3: Tam. 4, 5, or 6: Yusuf.
Not fair. Rosa $\frac{1}{2}$, Tam $\frac{1}{2} \cdot \frac{3}{6} = \frac{1}{4}$, Yusuf $\frac{1}{2} \cdot \frac{3}{6} = \frac{1}{4}$.

(e) Change method (b) to make it fair.
Example: Hearts: Rosa. Diamonds: Tam. Clubs: Yusuf. Spades: draw again.

At a school fair booth, you draw one card from a standard deck. You win \$6 if it is a heart. You pay \$2 for any other card.

(f) Find the expected value of what you win.
$6 \cdot \frac{13}{52} + (-2) \cdot \frac{39}{52} = \frac{78}{52} - \frac{78}{52} = 0$

(g) Is the game fair? Explain. Yes. The expected value is 0. On average, no one wins or loses.

At another booth, you flip 2 coins. You win \$5 if both are heads. You pay \$2 for anything else.

(h) Find the expected value. Is the game fair?
$5 \cdot \frac{1}{4} + (-2) \cdot \frac{3}{4} = \frac{5}{4} - \frac{6}{4} = -\frac{1}{4} = -\$0.25$. Not fair. You lose 25 cents a game, on average.

(i) You play this game 40 times. About how much money do you expect to win or lose in total?
$40 \times (-\$0.25) = -\$10$. You expect to lose about \$10.

(j) How much should you win for two heads to make the game fair?
$w \cdot \frac{1}{4} - \frac{6}{4} = 0$, so $w = 6$. You should win \$6.

7.3 - Expected Payoff

5. **I find the net gain for each outcome, and I find the expected payoff of a lottery, game, or plan.**

   The art club sells 250 raffle tickets for $4 each. One ticket wins $250. Five tickets win $30 each. The rest win nothing.

   (a) Fill in the table.

Outcome	Net gain x	$P(x)$	$x \cdot P(x)$
Win $250	$250 - 4 = 246$	$\frac{1}{250} = 0.004$	0.984
Win $30	$30 - 4 = 26$	$\frac{5}{250} = 0.02$	0.52
Win nothing	-4	$\frac{244}{250} = 0.976$	-3.904

   (b) Find the expected payoff. $0.984 + 0.52 - 3.904 = -\$2.40$

   (c) What does the expected payoff mean? On average, a buyer loses $2.40 per ticket.

   (d) The club sells all 250 tickets. How much money does the club raise?
   $250 \times \$2.40 = \$600$ (or $\$1{,}000 - \$250 - \$150 = \$600$)

   At another booth, you pay $3 to spin a spinner with 10 equal parts. 1 part wins $20. 2 parts win $5. The other 7 parts win nothing.

   (e) Find the net gain for each outcome.
   Win $20: $20 - 3 = 17$. Win $5: $5 - 3 = 2$. Win nothing: -3.

   (f) Find the expected payoff.
   $17(0.1) + 2(0.2) + (-3)(0.7) = 1.7 + 0.4 - 2.1 = 0$

   (g) Is the game fair? Should a player expect to win money?
   Yes, it is fair. No. On average, a player does not win or lose money.

   (h) The school wants to make $1 per player, on average. How much should a spin cost?
   Average prize: $20(0.1) + 5(0.2) = \$3$. Cost: $\$3 + \$1 = \$4$.
   Check: $16(0.1) + 1(0.2) + (-4)(0.7) = 1.6 + 0.2 - 2.8 = -1$

6. **I use expected value, and risk, to make a decision.**

A game console costs $350 to repair. The chance that it needs a repair in the first year is 0.06. Plan A is a one-year protection plan. It costs $29.

(a) Find the expected repair cost with no plan. $0.06 \cdot 350 = \$21$

(b) Which is cheaper on average: Plan A or no plan? By how much?
No plan. $\$29 - \$21 = \$8$ cheaper on average.

(c) Give a reason someone might still buy Plan A.
They could not pay a sudden $350 bill. The plan removes that risk.

(d) The store sells 500 of Plan A. About how much money does the store expect to make?
Each plan: $\$29 - \$21 = \$8$. $500 \times 8 = \$4{,}000$

Plan B costs $12. With Plan B, you still pay $100 if the console needs a repair.

(e) Find the expected cost of Plan B.
You always pay $12. You pay $100 more with probability 0.06.
$12 + 0.06 \cdot 100 = 12 + 6 = \$18$

(f) Which choice costs the least on average: no plan, Plan A, or Plan B?
Plan B ($18). Then no plan ($21). Then Plan A ($29).

(g) Mia has only $60 saved for surprise bills. Which choice would you tell her to pick? Explain.
Plan A. With no plan, she could get a $350 bill. With Plan B, she could get a $100 bill. Both are more than $60. With Plan A, she never pays more than $29.

(h) At what price would Plan A cost the same as no plan, on average?
$21, the expected repair cost with no plan.

Test-Style Questions

A test asks the same questions with more words. Read slowly. Circle the best answer.

7. The random variable X represents the number of times a randomly selected student rode the city bus last week. The table shows the probability distribution of X, but one probability is missing.

x	0	1	2	3	4
$P(x)$	0.42	0.25	0.18	?	0.05

What is the probability that a randomly selected student rode the city bus at least 2 times last week?

A 0.10 B 0.23 (C) 0.33 D 0.58

Missing: $1 - 0.90 = 0.10$. $P(X \geq 2) = 0.18 + 0.10 + 0.05 = 0.33$

8. Which of the following describes an empirical probability distribution?

A The probability of each sum when two fair number cubes are rolled, found by listing all 36 outcomes.

B The probability of each number of heads when 3 fair coins are flipped, found by listing all 8 outcomes.

(C) The probability of each number of goals a team scores, found by dividing the number of games with that many goals by the total number of games played last season.

D The probability of drawing each suit from a standard deck of 52 cards, found by counting the cards of each suit.

9. On each turn of a board game, a player spins a spinner and earns points. The random variable X represents the number of points a player earns on one turn. The table shows the probability distribution of X.

x	0	2	5	10
$P(x)$	0.4	0.3	0.2	0.1

What is the expected value of X?

A 0 (B) 2.6 C 4.25 D 17

$0(0.4) + 2(0.3) + 5(0.2) + 10(0.1) = 0 + 0.6 + 1.0 + 1.0 = 2.6$

10. In a game, a player rolls a fair number cube one time. If the cube lands on 6, the player wins $10. If the cube lands on any other number, the player loses $1. Which statement about the game is true?

A The game is fair, because the player wins on only one of the six numbers.
B The game is fair, because the expected value of the game is $0.
C The game is not fair, because the player loses about $0.83 per game, on average.
(D) The game is not fair, because the player wins about $0.83 per game, on average.

$10 \cdot \frac{1}{6} + (-1) \cdot \frac{5}{6} = \frac{10}{6} - \frac{5}{6} = \frac{5}{6} \approx \$0.83$

11. A company charges $75 for a one-year warranty on a washing machine. The company estimates that 4% of the machines will need a repair that costs $900 during the year, and that the other machines will need no repairs. Based on these estimates, what is the company's expected profit for each warranty it sells?

A $36 (B) $39 C $75 D $825

Expected repair cost: $0.04 \cdot 900 = \$36$. Profit: $\$75 - \$36 = \$39$

12. A school club sells 500 raffle tickets for $4 each. One ticket wins a $500 prize, and 5 tickets each win a $50 prize. All other tickets win nothing. What is the expected payoff for a person who buys one ticket?

A $-\$4.00$ (B) $-\$2.50$ C $\$1.50$ D $\$2.50$

$496 \cdot \frac{1}{500} + 46 \cdot \frac{5}{500} + (-4) \cdot \frac{494}{500} = 0.992 + 0.46 - 3.952 = -\$2.50$